PSP — ALL QUIZ & MCQ QUESTIONS, CHAPTER-WISE, WITH ANSWERS

Every MCQ ever captured for PSP, organised by IA-1 chapter, Question to Answer format, plus a full analysis of ALL quizzes (9 revision quizzes + 9 classroom quizzes + 4 quiz contests).

41 MCQs with answers — each code-trace answer was execution-verified; where the portal capture itself contained the official explanation, it is quoted. Companion: `PSP-ALL-CODING-QA.pdf` (100 coding questions with answers).

PART A — FULL QUIZ ANALYSIS (every quiz that exists)

A1. The 9 revision quizzes (graded, portal Quiz tab)

Quiz	Date	Your score	Chapter	Question text in hand?
Print function, Comments, Variable, Primitive, type()	2 Sep	0/12	CH2	Q1 of 6 captured; Q2-6 not
Python Input, Arithmetic operators, precedence	7 Sep	2/26	CH3	no text captured
Conditional Statements, if-else	9 Sep	8/8 (done)	CH4	no text (classroom version captured instead)
If-elif-else ladder	21 Sep	0/8	CH5	no text captured
Python Function, Nested if-else	23 Sep	0/14	CH6	no text captured
Function, Arguments, Return	28 Sep	8/8 done (2x)	CH8	no text captured
Iteration, Recursion, Range	12 Oct	0/14	CH10	Q1 of 5 captured; Q2-5 not
While loop	12 Oct	0/10	CH12	no text captured
Control flow, Break, Continue	16 Oct	0/8	CH11/12	no text captured

Analysis

- You solved 3/9. The three you solved (Conditional 8/8, Function+Return 8/8, Input partial) are your strongest; every unattempted quiz is a *trace* quiz (predict output / loop state) — exactly the IA-1 MCQ style.
- Highest-value gaps: **While loop (0/10)** and **Iteration/Recursion (0/14)** — these map straight to L10-L12, the last and most exam-weighted lectures. The captured classroom MCQs in this book (10 while-loop + 1 for/recursion) are the same style — drill those.
- Question text for unattempted quizzes only appears after starting the test (never started here). Use **View Solutions** on the portal if offered, or start them only when ready to spend the attempt.

A1b. Fetch status of ALL 9 revision quizzes — nothing skipped

#	Quiz (date)	Score	Chapter	Question text captured?	What to do
1	Print function… (2 Sep)	0/12	CH2	**Q1 of 6 only**	open id `7vc3w3or5w2v` → Q2–6
2	Input & operators… (7 Sep)	2/26	CH3	✅ **all 11 captured** (View Solutions)	see **B-EXTRA** B49–B59
3	Conditional Statements (9 Sep)	8/8	CH4	✅ **all 3 captured**	see **B-EXTRA** B42–B44
4	If-elif-else ladder (21 Sep)	0/8	CH5	**none**	open id `2q8icy0qbaby` (3 Qs)
5	Python Function, Nested if-else (23 Sep)	0/14	CH6	**none**	open id `eoyocgdhbvsm` (3 Qs)
6	Function, Arguments, Return (28 Sep)	8/8	CH8	✅ **all 4 captured**	see **B-EXTRA** B45–B48

#	Quiz (date)	Score	Chapter	Question text captured?	What to do
7	Iteration, Recursion, Range (12 Oct)	0/14	CH10	**Q1 of 5 only**	open id `6lvpokfzpgz5` (Q2–5)
8	While loop (12 Oct)	0/10	CH12	**none**	open id `sj0tck7nc5f2` (5 Qs)
9	Control flow, Break, Continue (16 Oct)	0/8	CH11/12	**none**	open id `pzuuhrktwqzn` (4 Qs)

Link format: `https://my.newtonschool.co/course/zqngjr790187/assessment/<id>`

Why 3 have no id: the tracker only ever opened the quizzes that showed instructions pages; the 3 attempted ones (rows 2, 3, 6) were never opened at all, so no URL was saved. They are reachable from the portal Quiz tab, and because they are attempted, **View Solutions should show their questions + correct answers**.

A2. The 9 classroom quizzes (in-class, learning-only, NOT graded) — all fully captured

Quiz	Where captured	Qs	Chapter	Status
Conditional Statements, if-else (classroom)	14 Sep window	3 MCQs	CH4	fully captured — Part B
If-elif-else ladder (classroom)	21 Sep window	3 MCQs	CH5	fully captured — Part B
Python Function, Nested if-else (classroom)	3 Oct window	3 MCQs	CH6	fully captured — Part B
if-else + logical operators (classroom)	5 Oct window	4 MCQs	CH9	fully captured — Part B
Input+operators+conditional (lab classroom)	lab window	3 MCQs	CH4/5	fully captured — Part B
Input, operators, datatypes (lab classroom)	lab window	10 MCQs	CH3	fully captured — Part B
Input+if basics (lab classroom)	lab window	3 MCQs	CH4	fully captured — Part B
While loop (classroom, batch window)	17 Oct window	5 MCQs	CH12	fully captured — Part B
While loop (classroom, lab window)	lab window	5 MCQs	CH12	fully captured — Part B

These 41 MCQs are the complete classroom-quiz bank -> Part B below.

A3. The 4 quiz contests (never opened — Start Test consumes an attempt)

PSP Contest 1, Wifi Demo, Demo_Contest, Mock Test — questions invisible until started. If wanted: portal -> PSP B -> Contest -> Quiz Contests, one attempt each.

PART B — ALL 41 MCQs, CHAPTER-WISE, Q -> A

Chapter 2 — Variables, Datatype, Typecasting, Print

B1. [CH2] PSP B Print function in python, Comments in python, Python Variable, Python Primitive

Print function in python, Comments in python, Python Variable, Python Primitive ... - Revision

QUESTION 1/6

What will be the data type of the variable value after executing the following Python code?

value = 50

value = "50"

print(type(value))

A

<class 'float'>

B

<class 'bool'>

C

<class 'int'>

D

<class 'str'>

ANSWER: <class 'str'>

Why: value is reassigned to "50" → str.

Chapter 3 — Input & Arithmetic Operators

B2. [CH3] Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

QUESTION 1/10

What is the output of the following Python expression?

print(8 // 4 ** 6 * 2 / 3 % 4 - 7)

A

-2

B

-7

-7.0

ANSWER: -7.0

Why: 8//4**6 → 0; 0*2/3 → 0.0; 0.0%4 → 0.0; 0.0-7 = -7.0 (official explanation in capture).

B3. [CH3] Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

QUESTION 2/10

What will be the output of the below given python code?

b = 10

a = 3

c = b < a

print(c)

A

True

False

ANSWER: False

Why: 10 < 3 is False.

B4. [CH3] Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

QUESTION 3/10

What will be the output of the below given python code?

num1 = 15

num2 = 28

num3 = 100

result1 = num1 <= num2

result2 = num3 > num1

print(result1 != result2)

A

True

False

ANSWER: False

Why: True != True → False.

B5. [CH3] Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

QUESTION 4/10

What will be the output of the below given python code?

x = 10

y = 5

result1 = (x >= 10)

result2 = (y == 5)

print(result1 == result2)

A

False

B

0

True

ANSWER: True

Why: True == True → True.

B6. [CH3] Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

QUESTION 5/10

What will be the output of the below given python code?

num1 = 5

num2 = 3

print(num1 > num2)

True

ANSWER: True

Why: 5 > 3 → True.

B7. [CH3] Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

QUESTION 6/10

What will be the output of the below given python code?

a = 2 * 3 // 5 % 6 ** 4

print(a)

A

0

1

ANSWER: 1

Why: 6**4=1296; 2*3//5=1; 1%1296=1.

B8. [CH3] Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

QUESTION 7/10

What will be the result of the following expression in Python?

```
x = 1
y = 2
z = 3
result = x == x <= x + y * z

print(result)
```

True

ANSWER: True

Why: x==x → True; True <= 7 → True (chained comparison).

B9. [CH3] Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

QUESTION 8/10

What will be the output of the below given python code?

```
num1 = 15
num2 = 28
result = num2 >= num1 + 5
print(result)
```

A

False

True

ANSWER: True

Why: 28 >= 20 → True.

B10. [CH3] Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

QUESTION 9/10

What will be the output of the following code if the user enters 10, 20 as input?

```
a, b = input().split(", ")
print(int(a) * int(b))
```

200

ANSWER: 200

Why: split(", ") → 10 * 20 = 200.

B11. [CH3] Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

QUESTION 10/10

What will be the output of the below given python code?

result = "apple,,banana,,cherry".split(",")

print(result)

['apple', '', 'banana', '', 'cherry']

ANSWER: ['apple', '', 'banana', '', 'cherry']

Why: double commas produce empty strings.

Chapter 4 — if-else

B12. [CH4] Classroom Quiz - Conditional Statements, Python if- else

QUESTION 1/3

Which of the following is the correct syntax for an if-else statement in Python?

A

```
if(condition)
  statements;
else:
  statements;
```

B

```
if condition
{
  statements
}
else
{
  statements
}
```

C

```
if condition:
  statements
elif condition:
  statements
else:
  statements
```

D

```
if condition:
  statements
else:
  statements
```

ANSWER: if condition: / else: — correct Python syntax

Why: Python needs a colon, no braces, no semicolons; C is a valid ladder but the question asks plain if-else — D is the correct if-else form.

B13. [CH4] Classroom Quiz - Conditional Statements, Python if- else

QUESTION 2/3

What will be the output of the following code?

```
x = 7
if x % 2 == 0:
    print(x*2)
else:
    print(x*3)
```

A

None of the Above

B

14

C

21

D

7

ANSWER: 21

Why: x=7 is odd → else branch → 7*3 = 21.

B14. [CH4] Classroom Quiz - Conditional Statements, Python if- else

QUESTION 3/3

What will be the output of the following Python code?

```
if False:
    print("1")
else:
    print("2")
```

ANSWER: 2

Why: if False → else → print("2").

B15. [CH4] Classroom Quiz - Python Input, Python operators, Conditional Statements, Input() function, Input(...

QUESTION 3/3

What will be the output of below code?

```
a = 10
b = 5
c = 15
result = 0
if a > b:
result = result+1
elif b < c:
result = result+2
if c > a:
result = result+3
print(result)
```

1

ANSWER: 4

Why: a>b → result=1 (elif skipped); c>a → result=4.

B16. [CH4] Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

QUESTION 1/3

What will be the output of the following code?

```
x = 20
if x >= 20:
  print("x is good")
else:
  print("x is not good")
```

A

x is less than 20

B

x is good

C

No output

D

x is 20 or more

ANSWER: x is good

Why: 20 >= 20 → if branch.

B17. [CH4] Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

QUESTION 2/3

What will be the output of the following code?

```
x = 125
if x > 50:
x = x - 50
  print(x)
if x > 150:
  x = x - 150
  print(x)
else:
  print(x)
```

A

25

75

B

100

75

C

75

75

D

125

75

ANSWER: 75 then 75 (two lines)

Why: 125>50 → x=75 print; 75>150 False → else print 75.

B18. [CH4] Classroom Quiz - Python Input, Python operators, Input() function, Input() for different datatype ...

QUESTION 3/3

What would be the output of the following code?

if 5 > 3:

print("Yes")

ANSWER: Yes

Why: 5>3 → print("Yes").

Chapter 5 — if-elif-else ladder

B19. [CH5] Classroom Quiz - If- elif-else ladder

QUESTION 1/3

Identify the correct elif ladder syntax in Python.

A

```
elif condition1:
  # block1
else:
  # block2
```

B

```
if condition1:
  # block1
elif condition2:
  # block2
```

C

```
if condition1:
  # block1
elif condition2:
  # block2
else:
  # block3
```

D

Both A and B

E

Both B and C

ANSWER: E — Both B and C

Why: A starts with elif (illegal alone); B (if/elif) and C (if/elif/else) are both valid.

B20. [CH5] Classroom Quiz - If- elif-else ladder

QUESTION 2/3

What will be the output of following code?

```
x = 10
if x > 0:
  print("block 1")
elif x <= 10:
  print("block 2")
elif x == 10:
  print("block 3")
else:
  print("block 4")
```

A

block 2

B

block 1

C

block 4

D

block 3

ANSWER: block 1

Why: x>0 is True → first branch prints; elif/else skipped.

B21. [CH5] Classroom Quiz - If- elif-else ladder

QUESTION 3/3

What will be the output of following code?

```
x = 10
if x > 0:
print("block 1")
if x <= 10:
print("block 2")
if x == 10:
print("block 3")
else:
print("block 4")
```

A

block 3

B

block 2

block 3

C

block 1

D

block 1

block 2

block 3

E

block 1

block 3

ANSWER: block 1 / block 2 / block 3 (three lines)

Why: Three independent ifs: x>0 yes, x<=10 yes, x==10 yes → block1, block2, block3.

B22. [CH5] Classroom Quiz - Python Input, Python operators, Conditional Statements, Input() function, Input(...

QUESTION 1/3

What will be the output of the following code?

```
x = 12
if x > 12:
print("Greater than 12")
elif x > 10:
print("Greater than 10")
elif x > 8:
print("Greater than 8")
else:
print("Less than or equal to 8")
```

Greater Than 10

Greater than 10

ANSWER: Greater than 10

Why: x=12: x>12 False → elif x>10 True → "Greater than 10" (exact casing).

B23. [CH5] Classroom Quiz - Python Input, Python operators, Conditional Statements, Input() function, Input(...

QUESTION 2/3

What will be output of below code if score = 75?

```
score = 75
if score >= 90:
print("A")
elif score >= 80:
print("B")
elif score >= 70:
print("C")
else:
print("D")
```

c

ANSWER: C

Why: 75 >= 70 → prints C (official explanation in capture).

Chapter 6 — Nested if-else

B24. [CH6] Classroom Quiz - Python Function, Nested if-else

QUESTION 1/3

What will the following code output?

```
x = 10
if x > 5:
if x < 15:
print("Yes")
else:
print("No")
```

Yes

ANSWER: Yes

Why: x=10: x>5 true, x<15 true → "Yes" (official explanation also in the capture).

B25. [CH6] Classroom Quiz - Python Function, Nested if-else

QUESTION 2/3

Neeraj is preparing for the finals. His three throws are recorded: throw1 = 86.7, throw2 = 90.6, and throw3 = 88.1. Based on his average throw, the coach gives feedback:

```
throw1 = 86.7
throw2 = 90.6
throw3 = 88.1

avg = (throw1 + throw2 + throw3) // 3

if avg >= 88:
if throw3 > throw2:
print("Excellent Finish")
elif avg < 89:
print("Great, but can be better")
else:
print("Superb")
else:
print("Needs Improvement")
```

What will be printed?

A

Excellent Finish

Great, but can be better

ANSWER: Great, but can be better

Why: avg = 265.4//3 = 88.0 → >=88; throw3(88.1) > throw2(90.6)? No → avg<89 yes → "Great, but can be better".

B26. [CH6] Classroom Quiz - Python Function, Nested if-else

QUESTION 3/3

What will be the output of following code?

```
loan = 80000
duration = 12
```

```
base_rate = 0.12
card_type = "Gold"

if card_type == "Silver":
rate = base_rate - 0.01
elif card_type == "Gold":
if duration > 12:
rate = base_rate - 0.01
else:
rate = base_rate - 0.02
elif card_type == "Platinum":
rate = base_rate - 0.03
else:
rate = base_rate

total_interest = loan * rate
total_payment = loan + total_interest

print(int(total_payment))
88000
```

ANSWER: 88000

Why: Gold + duration 12 (not >12) → rate = 0.12-0.02 = 0.10 → 80000+8000 = 88000.

Chapter 9 — Logical operators

B27. [CH9] Classroom Quiz - Python if- else, Python logical Operators

QUESTION 1/4

Which of the following is the incorrect syntax for using logical operators in an if statement?

A

```
if condition1 & condition2:
```

B

```
if condition1 or condition2:
```

C

```
if (condition1) and (condition2):
```

D

```
if condition1 and condition2:
```

ANSWER: A — if condition1 & condition2:

Why: & is the bitwise operator, not the logical operator; B, C, D are correct logical forms.

B28. [CH9] Classroom Quiz - Python if- else, Python logical Operators

QUESTION 2/4

What will be the output of the following code?

```
x = 20
y = 10
if not x > 15 and y > 15:
print("One condition is true")
```

A

No output

B

One condition is true

C

Error

D

None of these

ANSWER: No output

Why: not x>15 → not True = False; False and … → condition False → nothing printed.

B29. [CH9] Classroom Quiz - Python if- else, Python logical Operators

QUESTION 3/4

What would be the output of the following code?

print(5 > 3 and 3 < 2)

ANSWER: False

Why: 5>3 is True, 3<2 is False → True and False = False.

B30. [CH9] Classroom Quiz - Python if- else, Python logical Operators

QUESTION 4/4

What will be printed by this code?

x = 4

y = 10

if x < 5:

if not(y < 10 and x == 4):

print("Yes")

else:

print("Maybe")

else:

print("No")

ANSWER: Yes

Why: y<10 is False (y=10) → (False and True)=False → not False = True → inner if prints "Yes". Verified by running.

Chapter 10 — Recursive thinking (for/recursion)

B31. [CH10] PSP B Iteration in Python, Recursion Mechanism, Range () function, Intro and Types -

Iteration in Python, Recursion Mechanism, Range () function, Intro and Types - L ... - Revision

QUESTION 1/5

What will get printed during the last iteration of the following code?

x = 1

for i in range(1, 6):

if i % 2 == 0:

x *= -i

else:

x *= i

print(x)

ANSWER: 120

Why: x: 1 → -2 → -6 → 24 → 120; last iteration prints 120.

Chapter 12 — While loop + control flow

B32. [CH12] Classroom Quiz - While loop

QUESTION 1/5

What will get printed during the last iteration of the following code?

x = 1

while x < 100:

print(x)

x *= -3

A

81

B

-81

C

-243

D

243

ANSWER: -243

Why: prints 1,-3,9,-27,81,-243 then x=729 stops; last printed = -243.

B33. [CH12] Classroom Quiz - While loop

QUESTION 2/5

What will get printed during the last iteration of the following code?

x = 10

while x > 0:

print(x)

x -= 3

ANSWER: 1

Why: prints 10,7,4,1 then x=-2 stops; last = 1.

B34. [CH12] Classroom Quiz - While loop

QUESTION 3/5

What will get printed in the 5th iteration of the following code?

x = 1

while x <= 20:

print(x * x)

x += 2

ANSWER: 81

Why: 5th iteration x=9 → 9*9 = 81.

B35. [CH12] Classroom Quiz - While loop

QUESTION 4/5

What will be the final value of s?

```
n = 123
s = 0
while n > 0:
s += n % 10
n //= 10
```

ANSWER: 6

Why: s = 3+2+1 = 6.

B36. [CH12] Classroom Quiz - While loop

QUESTION 5/5

What will be the ouput of the below given code?

```
n = 123; s = 0
while n:
s += n % 10
n //= 10
print(n)
```

ANSWER: 0

Why: after the loop n=0; print(n) → 0.

B37. [CH12] Classroom Quiz - While loop

QUESTION 1/5

What will be printed in the 3rd iteration?

```
x = 2
while x <= 20:
print(x ** 2)
x += 6
```

ANSWER: 196

Why: x = 2,8,14,20 → squares 4,64,196,400; 3rd iteration prints 14*14 = 196.

B38. [CH12] Classroom Quiz - While loop

QUESTION 2/5

How many digits will be counted in the following code?

```
n = 98765
count = 0
while n > 0:
count += 1
n //= 10
```

ANSWER: 5

Why: 98765 has 5 digits → loop runs 5 times → count = 5.

B39. [CH12] Classroom Quiz - While loop

QUESTION 3/5

What will be printed in the last iteration?

```
x = 50
while x >= 0:
print(x // 10)
x -= 10
```

ANSWER: 0

Why: x = 50,40,…,0 → last print is 0//10 = 0.

B40. [CH12] Classroom Quiz - While loop

QUESTION 4/5

What will be the value of total after the loop ends?

```
x = 7
total = 0
while x <= 70:
if x % 10 == 7:
total += x
x += 1
```

ANSWER: 259

Why: 7+17+27+37+47+57+67 = 259 (verified by running).

B41. [CH12] Classroom Quiz - While loop

QUESTION 5/5

How many times does the loop run?

```
n = 1234
count = 0
while n > 0:
digit = n % 10
if digit % 2 == 0:
count += 1
n //= 10
```

ANSWER: 4

Why: digits 4,3,2,1 → the loop runs 4 times (count of even digits is 2, but the question asks loop runs).

B-EXTRA. Graded revision quizzes recovered via View Solutions (18 MCQs)

These three quizzes were already attempted, so the portal's **View Solutions** (read-only) showed every question with the official answer + explanation. Captured verbatim on 29 Oct.

Note: options are only partially visible in solution view (the chosen/correct option is shown, the rest are sometimes hidden) — question, correct answer and explanation are complete.

B42 — [CH4] Conditional Statements, if-else (9 Sep) — Q1/3

```
QUESTION 1/3

2/2

What would be the output of the following code?

if 10 < 5:

    print("Hello")

else:

    print("World")




World

Explanation:The condition 10 < 5 evaluates to False because 10 is not less than 5. Since the if block is not executed, the else block runs instead, producing the output: World.

Correct answer: World
```

B43 — [CH4] Conditional Statements, if-else (9 Sep) — Q2/3

```
QUESTION 2/3

2/2

What would be the output of the following code?

if 0:

    print("Zero is True")

else:

    print("Zero is False")




Zero is False

Explanation:In Python, the value 0 is considered falsy, meaning it evaluates to False in a boolean context. Therefore, the else block is executed, and the output is: Zero is False.

Correct answer: Zero is False
```

B44 — [CH4] Conditional Statements, if-else (9 Sep) — Q3/3

QUESTION 3/3

4/4

What will be the output of the following code?

```
x = 125
if x > 50:
   x = x - 50
   print(x)
if x > 150:
   x = x - 150
   print(x)
else:
   print(x)
```

A

25

75

75

75

Explanation:

x = 125 → Variable x is initialized with the value 125.

if x > 50: → Since 125 is greater than 50, this condition is True.

x = x - 50 → Subtract 50 from 125 → x becomes 75.

print(x) → Prints 75 (from the first if block).

if x > 150: → Now x is 75, which is not greater than 150, so this block is skipped.

else: → Since the above if is False, this else executes.

print(x) → Prints 75 again (from the else block).

Final Output:

75

75

C

100

75

D

125

75

CORRECT ANSWER

/course/zqngjr790187/assessment/dkuoquhedu0z

B45 — [CH8] Function, Arguments, Return (28 Sep) — Q1/4

QUESTION 1/4

2/2

You are given a function smart_calculator(a, b) that takes two integer numbers as input parameters and return a number based on the following condition

Returns Addition of the numbers if both are even

Returns the Multiplication of the numbers if both are odd

Returns the difference between the larger and smaller number if one is even and the other is odd

Input:

a = 4, b = 8

Output:

12

Explanation:

Both numbers are even → 4 + 8 = 12

What the function will return for the following input?

a = 3, b = 7

21

Explanation:Both 3 and 7 are odd numbers. So, the function returns their multiplication → 3 × 7 = 21. Final Output: 21

Correct answer: 21

B46 — [CH8] Function, Arguments, Return (28 Sep) — Q2/4

QUESTION 2/4

2/2

What will be the output of the following Python code?

```
def add(a, b):

    return a + b



print(add(2, 3) * 2)
```

10

Explanation:The function add(a, b) returns the sum of a and b. add(2, 3) returns 5. Then, 5 * 2 = 10. So the final result printed is 10.

B

8

C

6

D

Error

CORRECT ANSWER

B47 — [CH8] Function, Arguments, Return (28 Sep) — Q3/4

QUESTION 3/4

2/2

What will be the output of the following Python code?

```
def calculate(a, b, operator):
    if operator == '+':
        result = a + b
    elif operator == '-':
        result = a - b
    elif operator == '*':
        result = a * b
    elif operator == '/':
        result = a / b
    else:
        result = None

    print(result)


calculate(20, 20, '+')
```

40

Explanation:

a = 20, b = 20, and operator = '+'.

The function checks the operator using if-elif conditions.

Since the operator is '+', it performs 20 + 20.

result = 40 is calculated.

The function prints 40.

Output:

40

Correct answer: 40

B48 — [CH8] Function, Arguments, Return (28 Sep) — Q4/4

QUESTION 4/4

2/2

What will be the output of the following python code?

```
def multiply(x, y):
    return x * y


print(multiply(4, 2 + 1))
```

12

Explanation:The expression 2 + 1 is evaluated first, giving 3.

So the function call becomes multiply(4, 3).

The function returns 4 * 3 = 12.

Final Output:

12

B

8

C

Error

D

8 + 4

CORRECT ANSWER

/course/zqngjr790187/assessment/oocl3s62egkd

B49 — [CH3] Input & operators (7 Sep) — Q1/11

QUESTION 1/11

2/2

What will be the output of the below given python expression?

```
print(-8 % 3)
```

1

Explanation:

Let’s understand how the modulo operator % works with negative numbers in Python:

The modulo operator returns the remainder after division, but in Python, the result always has the **same sign as the divisor** (the number after the %).

Here, divisor is 3 (positive), so the result will be in the range 0 to 2.

Calculate: −8 ÷ 3 = −3 (floor division), because −3 ∗ 3 = −9 which is less than or equal to −8.

Remainder = −8 − (−9) = 1

✅ So, −8 % 3 evaluates to 1.

🖨️ Final Output: 1

Correct answer: 1

B50 — [CH3] Input & operators (7 Sep) — Q2/11

QUESTION 2/11

0/2

What will be the output of the below given python code?

```
print(-2 ** 2)
```

Explanation:

Let’s carefully evaluate this expression considering operator precedence:

Exponentiation (**) has higher precedence than the unary minus (-).

So, 2 ** 2 is evaluated first → 4.

Then the unary minus is applied: - (2 ** 2) = -4.

⚠️ Note: This is different from (-2) ** 2, which would be 4.

🖨️ Final Output: -4

Correct answer: -4

B51 — [CH3] Input & operators (7 Sep) — Q3/11

QUESTION 3/11

0/2

What will be the output of the below given python code?

```
print(-10 // 3)
```

Explanation:

Let’s understand how // (floor division) works with negative numbers in Python:

Normal division: -10 / 3 = -3.333...

Floor division // returns the greatest integer less than or equal to the actual result.

So, floor of -3.333... is -4.

✅ Python always rounds down (toward negative infinity) when using // with negative numbers.

🖨️ Final Output: -4

Correct answer: -4

B52 — [CH3] Input & operators (7 Sep) — Q4/11

QUESTION 4/11

0/2

What will be the output of the below given python code?

```
print(1 + 2 * 3 / 4 - 5 ** 2 % 6)
```

Explanation:

Let’s evaluate the expression step-by-step following Python's operator precedence rules:

Exponentiation **: → 5 ** 2 = 25

Multiplication and Division (left to right): → 2 * 3 = 6 → 6 / 4 = 1.5

Modulo %: → 25 % 6 = 1

Addition and Subtraction (left to right): → 1 + 1.5 = 2.5 → 2.5 - 1 = 1.5

✅ The expression combines arithmetic operations with different precedences, but Python handles them correctly according to standard rules.

🖨️ Final Output: 1.5

Correct answer: 1.5

B53 — [CH3] Input & operators (7 Sep) — Q5/11

QUESTION 5/11

0/2

What will be the output of the below given python expression?

```
print(5 + 3 * 2)
```

Explanation:

Let’s evaluate this expression using Python’s operator precedence rules:

Multiplication * has higher precedence than Addition +

So, 3 * 2 = 6 is evaluated first

Then, 5 + 6 = 11

✅ The expression follows the standard BODMAS/PEMDAS rules (multiplication before addition).

🖨️ Final Output: 11

Correct answer: 11

B54 — [CH3] Input & operators (7 Sep) — Q6/11

QUESTION 6/11

0/2

What will be the output of the following python code?

```
result = 3 * 4 + 5 / 2 - 1

print(result)
```

A

7.5

13.5

Explanation:

Let’s evaluate the expression step-by-step, following operator precedence:

Multiplication and Division first (left to right): → 3 * 4 = 12 → 5 / 2 = 2.5

Then Addition and Subtraction (left to right): → 12 + 2.5 = 14.5 → 14.5 - 1 = 13.5

✅ The final result is a float because division (/) in Python always returns a float.

🖨️ Final Output: 13.5

C

17

D

13

NOT ATTEMPTED

B55 — [CH3] Input & operators (7 Sep) — Q7/11

QUESTION 7/11

0/2

What will be the output of the following code if the user enters 10?

```
x = input("Enter a number: ")

y = int(x) + 5

print(type(y), y)
```

A

<class 'str'> 105

<class 'int'> 15

Explanation:

Let’s walk through what happens when the user enters 10:

x = input("Enter a number: ") → The user inputs 10 (as a string). So, x = "10"

y = int(x) + 5 → Converts "10" to integer → 10 → Adds 5 → y = 15

print(type(y), y) → type(y) is <class 'int'> → y is 15

📝 So, the type of y is int, and its value is 15.

✅ Final Output: <class 'int'> 15

C

<class 'str'> 15

D

Error, cannot convert string to int

NOT ATTEMPTED

B56 — [CH3] Input & operators (7 Sep) — Q8/11

QUESTION 8/11

0/2

What will be the output of the following python code?

```
x = 4
y = 2
z = 3
result = x ** y ** z // y % 5
print(result)
```

A

1

B

0

3

Explanation:

Let’s break down this expression step-by-step using Python's operator precedence:

Step 1: Exponentiation ** is right-associative

→ y ** z = 2 ** 3 = 8

→ x ** (y ** z) = 4 ** 8 = 65536

Step 2: Floor Division //

→ 65536 // 2 = 32768

Step 3: Modulo %

→ 32768 % 5 = 3

✅ The operations are applied in the correct order as per Python’s precedence rules.

🖨️ Final Output: 3

D

2

NOT ATTEMPTED

B57 — [CH3] Input & operators (7 Sep) — Q9/11

QUESTION 9/11

0/2

What will be the output of the following code?

```
x = 2

y = 3

z = 4

result = 10 + x * y ** z // (x + y) % z

print(result)
```

A

18

B

14

10

Explanation:

Let’s solve this step-by-step, respecting Python’s operator precedence:

Exponentiation ** comes first: → y ** z = 3 ** 4 = 81

Multiplication * comes next: → x * 81 = 2 * 81 = 162

Addition inside parentheses: → x + y = 2 + 3 = 5

Floor Division //: → 162 // 5 = 32 (floor division result)

Modulo %: → 32 % z = 32 % 4 = 0

Addition with 10: → 10 + 0 = 10

📌 Each step uses Python’s order of operations (PEMDAS), and intermediate values are calculated from inside out.

✅ Final Output: 10

D

12

NOT ATTEMPTED

B58 — [CH3] Input & operators (7 Sep) — Q10/11

QUESTION 10/11

0/4

What will be the output of the below given python code?

```
total_time = 185
hours = total_time // 60
minutes = total_time % 60
print(hours, minutes)
```

A

3 25

B

2 55

3 5

Explanation:

Explanation:

185 // 60 = 3 → 3 complete hours

185 % 60 = 5 → 5 remaining minutes

So, the final output is: 3 5

D

3 15

NOT ATTEMPTED

B59 — [CH3] Input & operators (7 Sep) — Q11/11

QUESTION 11/11

0/4

What will be the output of the below given python code?

```
P = 5000

R = 5

T = 3

A = P * ((1 + (R / 100)) ** T)

B = A - P

print(B)
```

788.125

Explanation:

Let’s understand what this code is doing — it’s calculating compound interest:

P = 5000: Principal amount

R = 5: Annual interest rate (in percent)

T = 3: Time (in years)

A: Total amount after compound interest → Using the formula: A = P * (1 + R/100) ** T → A = 5000 * (1 + 0.05) ** 3 = 5000 * (1.157625) = 5788.125

B = A - P: Compound Interest → 5788.125 - 5000 = 788.125

✅ The variable B stores the **compound interest** earned over 3 years.

🖨️ Final Output: 788.125

B

750.00

C

900.00

D

810.25

NOT ATTEMPTED

/course/zqngjr790187/assessment/3ddw7xc47w6v

B-CONTEST. MCQ PSP Contest 1 — all 10 questions (View Solutions)

Attempted 18/26, so the portal's read-only solutions view exposed every question.

B60 — [CH2–CH9] MCQ PSP Contest 1 (9 Oct) — Q1/10

```
QUESTION 1/10

2/2

Which output will the following code produce?

age = 16

has_id = True


if age >= 18:

    if has_id:

        print("Entry allowed")

    else:

        print("ID Required")

else:

    print("Underage")


A

Entry allowed

B

ID Required

Underage

Explanation:age = 16 → not ≥ 18 → outer if fails. Skips inner if, goes to else. Prints "Underage".

CORRECT ANSWER
```

B61 — [CH2–CH9] MCQ PSP Contest 1 (9 Oct) — Q2/10

```
QUESTION 2/10

0/2

What is the output of the following Code?

def square(x):

    return x * x


def add(x):

    return x + 5


def calculate(x):

    return square(x) + add(x)


print(calculate(3))


12

Correct answer: 17
```

B62 — [CH2–CH9] MCQ PSP Contest 1 (9 Oct) — Q3/10

QUESTION 3/10

2/2

What will be output of below code if score = 75?

```
score = 75

if score >= 90:
   print("A")
elif score >= 80:
  print("B")
elif score >= 70:
  print("C")
else:
  print("D")
```

C

Explanation:

Explanation:

score = 75

Condition 1: score >= 90 → False

Condition 2: score >= 80 → False

Condition 3: score >= 70 → True, so it prints C.

Remaining conditions are skipped after the first True.

Final Output: C

Correct answer: C

B63 — [CH2–CH9] MCQ PSP Contest 1 (9 Oct) — Q4/10

QUESTION 4/10

0/2

What will be the output of below code?

```
a = 10
b = 5
c = 15
result = 0
if a > b:
  result = result+1
elif b < c:
  result = result+2

if c > a:
  result = result+3

print(result)
```

6

Explanation:Given: a = 10, b = 5, c = 15 Step 1: if a > b → 10 > 5 → True → result += 1 Since the if was true, elif is skipped. Step 2: if c > a → 15 > 10 → True → result += 3 Final result: result = 4 print(result) prints: 4

Correct answer: 4

B64 — [CH2–CH9] MCQ PSP Contest 1 (9 Oct) — Q5/10

```
QUESTION 5/10

2/2

What will the following code output?

x = "False"

if x == True:

    y = 0

    if y == True:

        print(10)

    else:

        print(20)

else:

    print(30)




30

Explanation:

x = "False" → Here x is a string, not a boolean value.

In the condition if x == True:, Python checks whether the string "False" is equal to the boolean True.

This comparison is False because "False" (string) is not equal to True (boolean).

So, the control goes to the else part of the first if statement.

Hence, print(30) gets executed.

Correct answer: 30
```

B65 — [CH2–CH9] MCQ PSP Contest 1 (9 Oct) — Q6/10

QUESTION 6/10

2/2

What will be the output of the following Python print statement?

```
print("hello, welcome \nto newton school of technology")
```

A

```
Hello, welcome
 to Newton School of Technology
```

```
hello, welcome
to newton school of technology
```

C

```
hello, welcome \n to newton school of technology
```

D

```
hello, welcometo newton school of technology
```

E

```
hello, welcome
to Newton school of technology
```

CORRECT ANSWER

B66 — [CH2–CH9] MCQ PSP Contest 1 (9 Oct) — Q7/10

```
QUESTION 7/10

2/2

What will be the output of following code snippet?

x = 5

y = 15


if x > 4:

    print("True")

else:

    print(x)

    print("False")


True

B

15

False

C

Error

D

5

False

CORRECT ANSWER
```

B67 — [CH2–CH9] MCQ PSP Contest 1 (9 Oct) — Q8/10

```
QUESTION 8/10

6/6

What will be the output of following code snippet?

def check_loan(age, salary, credit_score, existing_loan):

    if age >= 21 and salary >= 30000:

        if (credit_score >= 750 and not existing_loan) or credit_score >= 800:

            return "Loan Approved"

        elif salary >= 50000 and credit_score >= 700:

            return "Manual Review"

        else:

            return "Loan Rejected"

    elif age >= 21 and salary < 30000:

        return "Low Salary"

    else:

        return "Age Not Eligible"


result = check_loan(25, 45000, 780, True)

print(result)


A

Loan Approved

B

Manual Review

Loan Rejected

D

Age Not Eligible

CORRECT ANSWER
```

B68 — [CH2–CH9] MCQ PSP Contest 1 (9 Oct) — Q9/10

QUESTION 9/10

0/4

What will be the output of the following code?

```
x = 125
if x > 50:
   x = x - 50
   print(x)
if x > 150:
   x = x - 150
   print(x)
else:
   print(x)
```

A

25

75

75

75

Explanation:

x = 125 → Variable x is initialized with the value 125.

if x > 50: → Since 125 is greater than 50, this condition is True.

x = x - 50 → Subtract 50 from 125 → x becomes 75.

print(x) → Prints 75 (from the first if block).

if x > 150: → Now x is 75, which is not greater than 150, so this block is skipped.

else: → Since the above if is False, this else executes.

print(x) → Prints 75 again (from the else block).

Final Output:

75

75

C

100

75

125

75

INCORRECT ANSWER

B69 — [CH2–CH9] MCQ PSP Contest 1 (9 Oct) — Q10/10

QUESTION 10/10

2/2

What will be the output of the following code?

```
x = 17


if x < 10:
    x = x + 10
    print(x)
elif x < 20:
    if x % 2 == 0:
        x = x + 2*x
        print(x)
    else:
        x = x + 20
        print(x)
else:
    print(x)
```

37

Explanation:

The value of x is initially 17.

The first if condition x < 10 is false, so we skip that block.

The elif x < 20 condition is true since 17 < 20.

Now, inside this block, we check if x % 2 == 1. Since 17 is odd, this condition is false.

So we update x as: x = x + 20 → x = 17 + 20 → x = 37.

Then we print the updated value of x, which is 37.

Output: 37

Correct answer: 37

Newton School

PART C — NOT FETCHED (chapter-wise) + IS THIS SUFFICIENT?

C1. Chapter-wise: what was asked/shared but could NOT be fetched

Chapter	Missing item	How to get it yourself	Priority for IA-1
CH1 Flowcharts	**Zero questions in any capture** (Flowchart was in the Contest-1 syllabus; the Contest-1 paper itself was never captured)	Ask a classmate who wrote Contest 1; or draw 3 yourself (login, even/odd, sum till N)	**MEDIUM** — the one chapter with no practice; theory only, low marks risk
CH2 Print/Variables	Revision quiz *Print function...* Q2-6 (5 MCQs)	My Assessments -> that quiz -> View Solutions	LOW — 5 coding + 1 MCQ in hand
CH3 Input/Arithmetic	**Input & operators quiz (7 Sep, 2/26) full text**; Contest-1 Qs	Quiz tab → that quiz → View Solutions; Contest-1 paper from a peer	LOW-MEDIUM — you already have 12 coding + 10 MCQs here
CH4 if-else	**Conditional Statements quiz (9 Sep, you scored 8/8 — text not saved)**; Contest-1 Qs	Quiz tab → View Solutions; Contest-1 paper from a peer	LOW — 11 coding + 7 MCQs in hand
CH5 if-elif ladder	Revision quiz *If-elif ladder* (0/8) full text	View Solutions / attempt	**MEDIUM** — 12 coding + 5 MCQs in hand, but 0/8 on the graded quiz
CH6 Nested if-else	Revision quiz *Python Function, Nested if-else* (0/14) full text	View Solutions / attempt	**MEDIUM** — 7 coding + 3 MCQs + workbook in hand
CH7 Functions	nothing missing (workbook 47 Qs covers it)	—	SUFFICIENT
CH8 Function+if-else	**Function, Arguments, Return quiz (28 Sep, 8/8 — text not saved)**	Quiz tab → View Solutions	LOW — already aced it
CH9 Logical	nothing	—	SUFFICIENT (9 coding + 4 MCQs)
CH10 Recursion	Revision quiz *Iteration/Recursion* Q2-5 (4 MCQs)	View Solutions / attempt	**MEDIUM-HIGH** — only 4 coding + 1 MCQ + workbook Unit 12 in hand; recursion is hard-exam material
CH11 For loop	**97 Codeforces statements** (63 if-else/math + 34 loops) + 12-code list — titles only	Open each CF link (2/day directive); sheets linked in #psp-26	MEDIUM for breadth; the 11 portal coding Qs are exam-style, CF is extra practice
CH12 While loop	Revision quiz *While loop* (0/10) + *Control flow* (0/8) full text	View Solutions / attempt	**MEDIUM** — but you have 15 coding + 10 MCQs here (biggest bank in the book)
ALL	**4 quiz-contest tests** (PSP Contest 1, Wifi Demo, Demo_Contest, Mock Test); **2 Google Sheets**; anything posted in Slack after 23 Oct (session expired)	Contest tab (one attempt each); Sheets: File -> Make a copy; Slack: QR re-login	LOW-MEDIUM — Contest 1 doubles as a past IA-1-style paper

C2. Is this enough to handle HARDER questions on exam day?

Verdict: YES for every chapter except CH1 (flowcharts) and partially CH10 (recursion) — with two caveats.

- **Why it is enough:** the exam can only be built from L1-L12. For CH2-CH9 and CH11-CH12 you now hold *every question the course ever set* (100 coding + 41 classroom MCQs + 47 workbook), all with answers. Harder questions are the same patterns with nastier numbers — the hardest portal items (Newton and Rounding [hard], Check if Rare, Grade Report) are solved in the coding book. You have seen every trap the course uses: exact output casing, XOR ("but not both"), 10^18 ranges, break-early, 9->0 digit wrap, round-half-down.
- **Caveat 1 — CH1:** you have *zero* flowchart questions. Risk is small (usually 1 theory Q: shapes, arrows, diamond = decision) — revise the CH1 notes in `PSP-IA1-KIT.pdf` Part 1 and draw 3 by hand; nothing else to fetch.
- **Caveat 2 — graded-quiz style (CH5, CH6, CH10-12):** the *graded* revision quizzes you never opened are the only place the examiner's own harder MCQ wording exists. If IA-1 reuses that style, your practice base for recursion traces is thinner (4

coding + 1 MCQ + 5 workbook). **Do the unfetched quiz questions via View Solutions if the portal allows it** — that closes the last real gap (CH5: 8 Qs, CH6: 14 Qs, CH10: 4 Qs, CH12: 18 Qs).

- **Not needed for passing:** the 97 Codeforces statements — breadth practice only, "2/day" ongoing work, not IA-1 material. Do them after the exam; skip tonight.

Bottom line: read `PSP-IA1-KIT.pdf` (concepts + protocol), drill this MCQ book + the coding book chapter-wise, open View Solutions on the graded quizzes you scored 0 on (CH5, CH6, CH10, CH12) if available — sufficient everywhere else.

PART D — EVERY QUIZ / CONTEST ON THE PORTAL (master inventory)

Nothing is omitted here: all 22 assessments that exist on the PSP portal are listed, with what text exists for each.

D1. Master inventory

#	Type	Name	Date	Qs	Total XP	Your score	Question text
1	Revision	Print function, Comments, Variables, Primitives	2 Sep	6	12	0/12	Q1 only (see D2)
2	Revision	Python Input, operators, precedence	7 Sep	11	26	2/26	✅ all 11 → B49–B59
3	Revision	Conditional Statements, if-else	9 Sep	3	8	8/8	✅ all 3 → B42–B44
4	Revision	If-elif-else ladder	21 Sep	3	8	0/8	✗ (Start Test only)
5	Revision	Python Function, Nested if-else	23 Sep	3	14	0/14	✗ (Start Test only)
6	Revision	Function, Arguments, Return	28 Sep	4	8	8/8	✅ all 4 → B45–B48
7	Revision	Iteration, Recursion, Range	12 Oct	5	14	0/14	Q1 only (see D2)
8	Revision	While loop	12 Oct	5	10	0/10	✗ (Start Test only)
9	Revision	Control flow, Break, Continue	16 Oct	4	8	0/8	✗ (Start Test only)
10–18	In-class	9 classroom quizzes	Sep–Oct	41	—	mixed	✅ all 41 → Part B
19	Contest	MCQ PSP Contest 1	9 Oct	10	26	18/26	✅ all 10 → B60–B69
20	Contest	Wifi Demo	2 Sep	2	4	0/4	✗ instructions deleted on portal
21	Contest	Demo_Contest	28 Aug	2	4	2/4	✗ instructions deleted on portal
22	Contest	Mock Test	27 Aug	5	10	0/10	✗ instructions deleted on portal

Totals: 104 MCQs exist (44 revision + 41 in-class + 19 contest). **69 are in this book** (41 + 18 + 10). Contest question counts for rows 20–22 are derived from their XP (2 XP/question), not from a visible page.

D2. Visible pre-start previews (not part of the 69)

Print function quiz (2 Sep) — Q1 of 6 (full options visible)

```
What will be the data type of the variable value after executing the following Python code?


value = 50

value = "50"

print(type(value))


A  <class 'float'>

B  <class 'bool'>

C  <class 'int'>

D  <class 'str'>
```

Iteration / Recursion / Range quiz (12 Oct) — Q1 of 5

```
What will get printed during the last iteration of the following code?

x = 1

...
```

D3. Contest assessment IDs (for your manual access)

Contest	Assessment ID	Direct link
MCQ PSP Contest 1	`01kpey3c9w7q`	https://my.newtonschool.co/course/zqngjr790187/assessment/01kpey3c9w7q
Wifi Demo	`c28pfi9sl6nv`	`…/assessment/c28pfi9sl6nv` (dead — "could not fetch instructions")
Demo_Contest	`g0augn5vh7fk`	`…/assessment/g0augn5vh7fk` (dead)
Mock Test	`41pvskqaxg02`	`…/assessment/41pvskqaxg02` (dead)